

ABHISHEK SURESH

551-229-8434 as18757@nyu.edu [LinkedIn](#) [GitHub](#)

EDUCATION

Master of Science, Information Systems

New York University, New York, USA

Sep 2023 – May 2025

Bachelor of Technology, Computer Science & Engineering

SRM Institute of Science and Technology, Chennai, India

May 2015 – May 2019

EXPERIENCE

AI Engineer

Aug 2025 – Present

Archemy Inc, New York, USA

- Designed a **serverless RAG pipeline** using **SageMaker Serverless endpoint (hugging face embedding all-MiniLM-L6-v2 model)** ingesting vector representation in **AWS S3 Vectors** with **API Gateway and Lambda** trigger, enabling **sub-3-second semantic search** at **90% lower cost** than managed vector DB alternatives
- Engineered a **SQL-to-vector knowledge graph pipeline** syncing MySQL ontology tables into **ChromaDB** backed by a **NetworkX** graph (648 nodes, 1,200+ edges); fine-tuned **MiniLM-L6-v2** embeddings on domain-specific contrastive pairs, improving **semantic search accuracy by 70%**
- Built an **LLM evaluation framework** with **LangSmith** and **RAGAS** tracking retrieval precision, answer faithfulness, and latency; automated feedback loops flagged low-confidence outputs for human review, improving RAG reliability by **30%**
- Architected and provisioned an **Aurora Serverless v2 PostgreSQL database via Terraform**, designing normalized schemas for audit trails and operational data powering an internal tooling platform; accessed via RDS Data API to eliminate connection overhead in a serverless environment

RPA Developer

Jan 2023 – Aug 2023

Bread Financial, India

- Led design and delivery of **100+ enterprise UiPath bots** in production using Orchestrator, queues, triggers, and unattended runners; mentored junior developers and enforced **REFramework coding standards** and SLA compliance
- Integrated **AWS SageMaker REST APIs** into UiPath workflows enabling real-time fraud detection and risk scoring; designed multi-source data reasoning pipeline combining anomaly signals, SQL outputs, and REST feeds improving fraud recall by **18%**
- Implemented **CI/CD-style bot promotion** across DEV, QA, and PROD environments using version-controlled packages and Orchestrator release governance, aligning with SDLC best practices
- Established centralized bot monitoring and alert-based escalation workflows; drove cross-functional collaboration across 3+ stakeholder teams reducing manual intervention by **20%**

Systems Engineer

Sep 2019 – Jan 2022

Infosys Ltd, India

- Developed using **Automation Anywhere** Control Room integrating **ML model inference** via REST APIs for anomaly detection and event-driven routing across IT operations; designed **checkpoint-based recovery** and workload balancing strategies reducing failure recovery time by **20 seconds** per incident
- Designed **checkpoint-based recovery** and workload balancing strategies reducing failure recovery time by **20 seconds** per incident
- Built **ML model deployment pipelines** using **Docker** and **GitHub Actions CI/CD**, automating model packaging and promotion across dev, staging, and production
- Designed data validation pipelines using **airflow** feeding validated predictions into **RPA orchestration layers**, cutting pipeline failure rate by **30%**

PROJECTS

Autonomous Financial Research Platform | AWS · OpenAI Agents SDK · Playwright MCP · Terraform

[Link](#)

- Built an **end-to-end agentic platform** ECS-hosted researcher agent browses live market data and **SEC EDGAR** via **Playwright MCP**; multi-agent **SQS pipeline** (Tagger + Reporter) runs autonomously every 2 hours via **EventBridge**, surfacing structured investment reports via **Next.js**, reducing research turnaround from hours to **under 60 seconds** across **12 AWS services** managed via Terraform

MetaML Knowledge Graph Dashboard | Streamlit · Knowledge Graph · SQL-to-Ontology Pipelines · FastAPI · Vercel

[Link](#)

- Built an interactive Streamlit dashboard for MetaML taxonomy exploration over a three-level ontology (domains, dimensions, and areas), with five modules that include **semantic search**, **knowledge graph navigation**, **concept similarity**, **free-text classification**, and **coverage/overlap auditing serving** as the primary interface for taxonomy discovery and validation

Product Price Prediction using LLMs | Python, Hugging Face, LoRA, QLoRA, Ollama

[Link](#)

- Fine-tuned **GPT-4**, **Claude**, and **LLaMA 3.1** on a **400K** Amazon product dataset using **LoRA** and **QLoRA** for parameter-efficient adaptation; designed **prompt engineering templates** and **evaluation benchmarks** (MSLE, MAE) improving pricing accuracy by **20%** over classical ML baselines.

SKILLS

- AI / Agents** : OpenAI Agents SDK, LangGraph, LangChain, MCP, Playwright, Autonomous Agents, Prompt Engineering, Guardrails, Tool Use
- RAG / Vector Search** : ChromaDB, AWS S3 Vectors, BM25, MMR Reranking, Knowledge Graphs, SageMaker Serverless Embeddings
- Evaluation & MLOps** : LangSmith, RAGAS, SageMaker, Airflow, Docker, Terraform, Model Drift Monitoring, LLM Evaluation Frameworks
- Cloud & Data** : AWS (Bedrock, ECS Fargate, SageMaker, S3 Vectors, Aurora Serverless, Lambda, API Gateway, SQS, EventBridge), GCP, Azure ML, Python, SQL, MongoDB, PostgreSQL